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Dear Editors,

On behalf of all co-authors (Rodrigo Almeida de Oliveira, Juliano de Paulo Ribeiro, and Prof. Edson Emilio Scalabrin), I am pleased to submit our manuscript titled **“Process Mining and Stochastic Modeling for Software Process Forecasting: A Systematic Mapping Study with an LLM-Assisted Protocol”** for consideration in *Information and Software Technology*.

Novelty and contribution. This manuscript reports a systematic mapping study at the intersection of process mining (PM), stochastic modeling, and machine learning applied to software process forecasting. The contribution is threefold:

1. **Methodological:** we operationalise an LLM-assisted screening protocol with cross-model verification (Claude Haiku 4.5 as primary screener, Claude Sonnet 4.6 as verifier), achieving substantial Cohen’s κ (binary $\kappa = 0.695$ for title/abstract, 0.694 for full-text) on a stratified random 20% LLM–LLM sample. Independent human double-screening ($n = 468$ T/A, $n = 177$ FT) showed fair/slight agreement and is reported as a limitation. The protocol extends the recent work of Khraisha et al. (2024) and Syriani et al. (2023) to the software-process-mining domain at a scale of 6,147 papers screened across two corpus tiers (5,783 after cross-source deduplication plus 364 control and manually imported items).
2. **Empirical:** we confirm 340 distinct primary studies after cross-tier deduplication (404 raw confirmations: 169 working-set + 212 + 23 auxiliary) covering 1995–2026. All 169 working-set studies were scored against an 8-criterion reporting-quality rubric (Dybå and Dingsøyr, 2008; 136 with $QA \geq 4/8$). Combined-tier findings use the 318-study analytical subset (259 with $QA \geq 4/8$).
3. **Synthetic:** we report five empirically grounded findings (F1–F5), propose the SPMF taxonomy (Software Process Mining and Forecasting) along three orthogonal dimensions, and articulate a four-direction research agenda (RA1–RA4). Only 1 of 340 papers jointly satisfies $TF\text{-}PM \cap TF\text{-}ST \cap TF\text{-}FC$ — evidence that the mapped gap is structural rather than an artefact of the search queries. PATHCAST is sketched as a future-work hypothesis, not as an empirically validated pipeline of this mapping study.

Fit with IST scope. The manuscript squarely fits IST’s scope on empirical software engineering methods, mining software repositories, and process modeling. It complements the journal’s growing body of evidence-based secondary studies while contributing to the methodological literature on LLM-assisted reviews.

Reproducibility. A complete replication package is openly available at Zenodo (DOI 10.5281/zen-

odo.21939600), including all search strings, screening prompts, JSON schemas, decision CSVs at every pipeline stage, the 340-study included list (`included_studies_340.csv`), the cross-model κ scripts, and the structured extraction tables.

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Prior submission and conflicts. The manuscript is original work, has not been published elsewhere, and is not under consideration by any other journal. All authors declare no competing financial interests. The use of generative AI tools in the research is declared in the manuscript per Elsevier policy. AI tools are *not* listed as authors and bear no responsibility for editorial judgements.

Suggested reviewers.

- **Wil M. P. van der Aalst** — RWTH Aachen University, foundational PM researcher (`wvdaalst@pads.rwth-aachen.de`)
- **Maria Teresa Baldassarre** — University of Bari, empirical SE and secondary studies
- **Eugene Syriani** — University of Montreal, LLM-assisted SLR methodology
- **Tore Dybå** — SINTEF / University of Oslo, evidence-based SE

Reviewers to avoid. None on basis of co-authorship within the past five years. Specific exclusions: thesis advisors and committee members at PUCPR.

Sincerely,

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